

Hypothesis Testing Report: Correlation Between Democracy Level And Gender Representation in Media

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Abstract—This report tests whether country-level democracy is associated with female representation in diplomatic imagery. Using image-based person detections, a country-year panel is assembled and a female representation ratio is derived. The panel is merged with V-Dem democracy indicators, and Spearman’s rank correlation is computed at the country level and significance is assessed via a permutation test. A statistically significant positive monotonic relationship is observed, although the estimated effect size is moderate.

I. DATA SOURCES

The diplomatic news and imagery data used in this study are publicly provided by the VRL Lab [1] originated from the GlobalDiplomacyNet project [2]. We use the Diplomatic Text Dataset hosted on Harvard Dataverse [3]. Democracy indicators are taken from the V-Dem Country-Year dataset (Full+Others, v15) [4].

II. DATA AND VARIABLE CONSTRUCTION

A. Extracting Female Ratio – Country Panel from Diplomatic Imagery

For each country folder, `news.jsonl` and `images.jsonl` are used. Each image record is matched to a news record via `news-id` (in `images.jsonl`) and `id` (in `news.jsonl`). The year is parsed from the `date` field in `news.jsonl`.

For each image, the number of detected people is defined as:

$$people_total = male_count + female_count. \quad (1)$$

Images with $people_total \leq 0$ are excluded because the female representation ratio is undefined when no people are detected, and including such images would mechanically bias aggregate ratios toward zero without providing information about gender composition.

Detected counts are aggregated to the country-year level and the main outcome is defined as:

$$female_ratio_{c,t} = \frac{female_total_{c,t}}{people_total_{c,t}}. \quad (2)$$

B. Merge with Democracy Indicators

Constructed country-year panel is merged with V-Dem indicators using $(country, year)$. The primary democracy measure in this report is `v2x_polyarchy` [4].

TABLE I
SUMMARY OF CONSTRUCTED PANEL, MERGE COVERAGE, AND TEST SAMPLE (POST-2000).

Statistic	Value
Country-year observations (all years)	1359
Countries observed (post-2000 panel)	143
Countries used in hypothesis test (post-2000)	137
Countries dropped (no V-Dem match post-2000)	6
Years covered (post-2000)	2000–2024
Share with $people_total > 0$	1.00
V-Dem match rate (<code>v2x_polyarchy</code> not missing)	95.36%

The post-2000 panel contains 143 countries; however, 6 countries have no matched V-Dem observations in the post-2000 period and are excluded from the hypothesis test, resulting in an analysis sample of 137 countries.

C. Data Coverage and Quality Checks

Several checks are performed to characterize coverage and potential sources of measurement noise in the constructed panel.

Coverage over time: The dataset contains extremely sparse pre-2000 coverage (Table II). Because these early observations are based on very small sample sizes, they can induce visually misleading long-run patterns in aggregated time series (Fig. 1).

People detection filter: Images with $people_total \leq 0$ are excluded because the female representation ratio is undefined when no individuals are detected. Including such images would mechanically push aggregate ratios toward zero without providing information about gender composition.

V-Dem merge success: The country-year panel is merged to V-Dem using $(country, year)$. The merge quality is summarized by the share of constructed country-year rows with non-missing `v2x_polyarchy`. In the current sample, the match rate is approximately 95%, indicating that most country-year observations can be linked to democracy indicators.

III. RESEARCH QUESTION AND HYPOTHESES

Research question: *Is there a positive relationship between the level of democracy and the visual representation of women in diplomacy-related media across countries?*

We test for a monotonic association using Spearman's rank correlation:

- H_0 : There is no relationship between democracy levels and the visual representation of women in diplomacy across countries. (Spearman $\rho = 0$).
- H_1 : Countries with higher levels of democracy tend to have a greater visual representation of women in diplomacy across countries. (Spearman $\rho \neq 0$).

We use a two-sided test with $\alpha = 0.05$.

IV. METHOD AND HYPOTHESIS TESTING

TABLE II
PRE-2000 DATA COVERAGE IN THE CONSTRUCTED COUNTRY-YEAR PANEL.

Year	$n_countries$	Rows	$people_total$	n_images_people
1950	1	1	10	1
1970	1	1	10	2
1998	1	1	1	1
1999	1	1	27	9

A. Sample Restriction (Post-2000)

Hypothesis testing is conducted using observations from 2000 onward. Pre-2000 coverage in the constructed panel is extremely sparse and uneven (Table II), which can yield unstable ratios and misleading long-run patterns. Restricting to the post-2000 period improves temporal comparability and ensures that country-level aggregates are not driven by isolated early-year artifacts.

B. Country-Level Aggregation

The hypothesis test is conducted at the country level. Using post-2000 country-year observations, we construct a single female representation measure per country:

$$female_ratio_c = \frac{\sum_{t \geq 2000} female_total_{c,t}}{\sum_{t \geq 2000} people_total_{c,t}}. \quad (3)$$

This aggregation implicitly weights years by the number of detected people, reducing sensitivity to sparse country-year cells.

Democracy is summarized as the mean polyarchy score over the same period:

$$polyarchy_c = \frac{1}{T_c} \sum_{t \geq 2000} v2x_polyarchy_{c,t}. \quad (4)$$

C. Spearman Correlation and Permutation Test

We compute Spearman's rank correlation ρ between $female_ratio_c$ and $polyarchy_c$ across countries ($N = 137$). Statistical significance is assessed using a permutation test. Under the null hypothesis of no association, we repeatedly

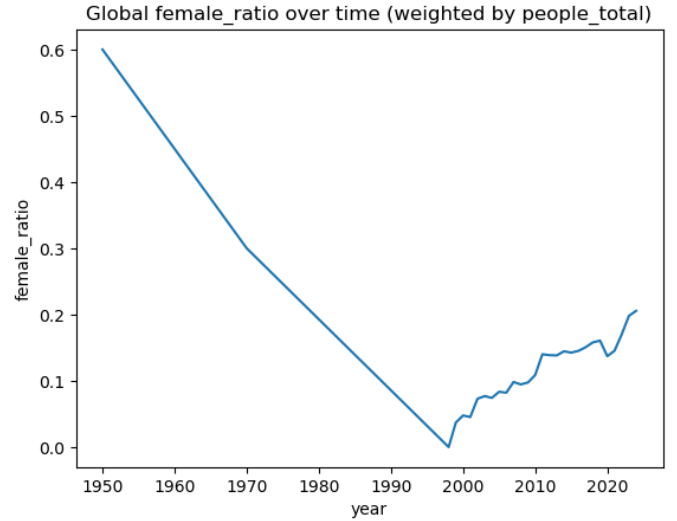


Fig. 1. Global female representation ratio over time, totaled by year and weighted by $people_total$. The apparent pre-2000 pattern is driven by extremely sparse coverage (Table II).

shuffle $polyarchy_c$ across countries, recompute ρ for each shuffle, and estimate a two-sided permutation p-value as:

$$p_{\text{perm}} = \frac{1 + \sum_{b=1}^B \mathbb{I}(|\rho_b| \geq |\rho_{\text{obs}}|)}{B + 1}. \quad (5)$$

V. RESULTS

At the country level ($N = 137$), the estimated Spearman correlation indicates a positive monotonic association:

$$\rho = 0.4959, \quad p_{\text{perm}} = 1.0 \times 10^{-4}. \quad (6)$$

Since $p_{\text{perm}} < 0.05$, we reject H_0 and conclude that countries with higher democracy scores tend to exhibit higher female representation in diplomacy-related imagery.

A. Effect Size Interpretation

The magnitude $\rho \approx 0.50$ suggests a moderate cross-country association. Substantively, this indicates that democracy is meaningfully related to female representation, though substantial variation remains unexplained by democracy alone.

VI. EXPLORATORY VISUAL EVIDENCE

To contextualize the hypothesis test and assess data structure, following exploratory plots are generated:

- Global time trend of $female_ratio$ aggregated by year (Fig. 1),
- Country-level scatter of female representation versus democracy (Fig. 2),
- Year-by-year Spearman correlation between country-year $female_ratio$ and $v2x_polyarchy$ (Fig. 3).

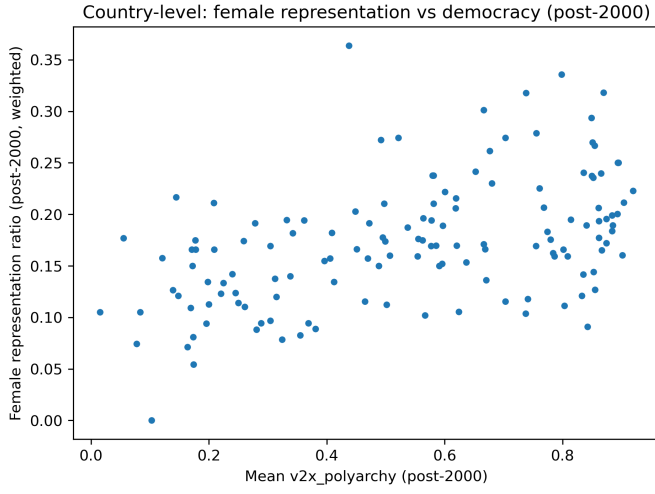


Fig. 2. Country-level association between female representation and democracy. Each point is a country, where female representation is aggregated as $\sum_{t \geq 2000} \text{female_total} / \sum_{t \geq 2000} \text{people_total}$ and democracy is the mean $v2x_polyarchy$ over 2000–2024.

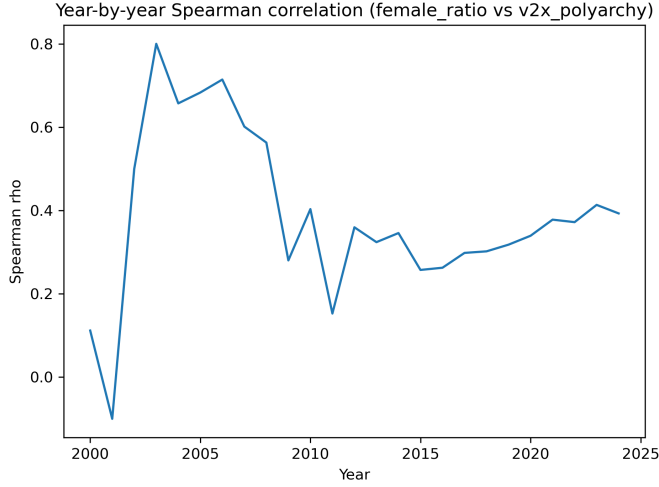


Fig. 3. Exploratory year-by-year Spearman correlation between country-year female representation and $v2x_polyarchy$ (post-2000).

VII. LIMITATIONS

- **Aggregation choice:** Country-level aggregation compresses within-country temporal dynamics; the test captures cross-country association rather than within-country change over time.
- **Remaining confounding:** Democracy may correlate with other structural factors (e.g., media practices, diplomatic visibility, region, development) that also influence representation; correlation does not imply causation.

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- [4] “V-dem dataset (country-year: Full+others, v15),” Varieties of Democracy (V-Dem) Institute, 2025, accessed 2026-01-07. [Online]. Available: <https://www.v-dem.net/data/the-v-dem-dataset/country-year-v-dem-fullothers-v15/>